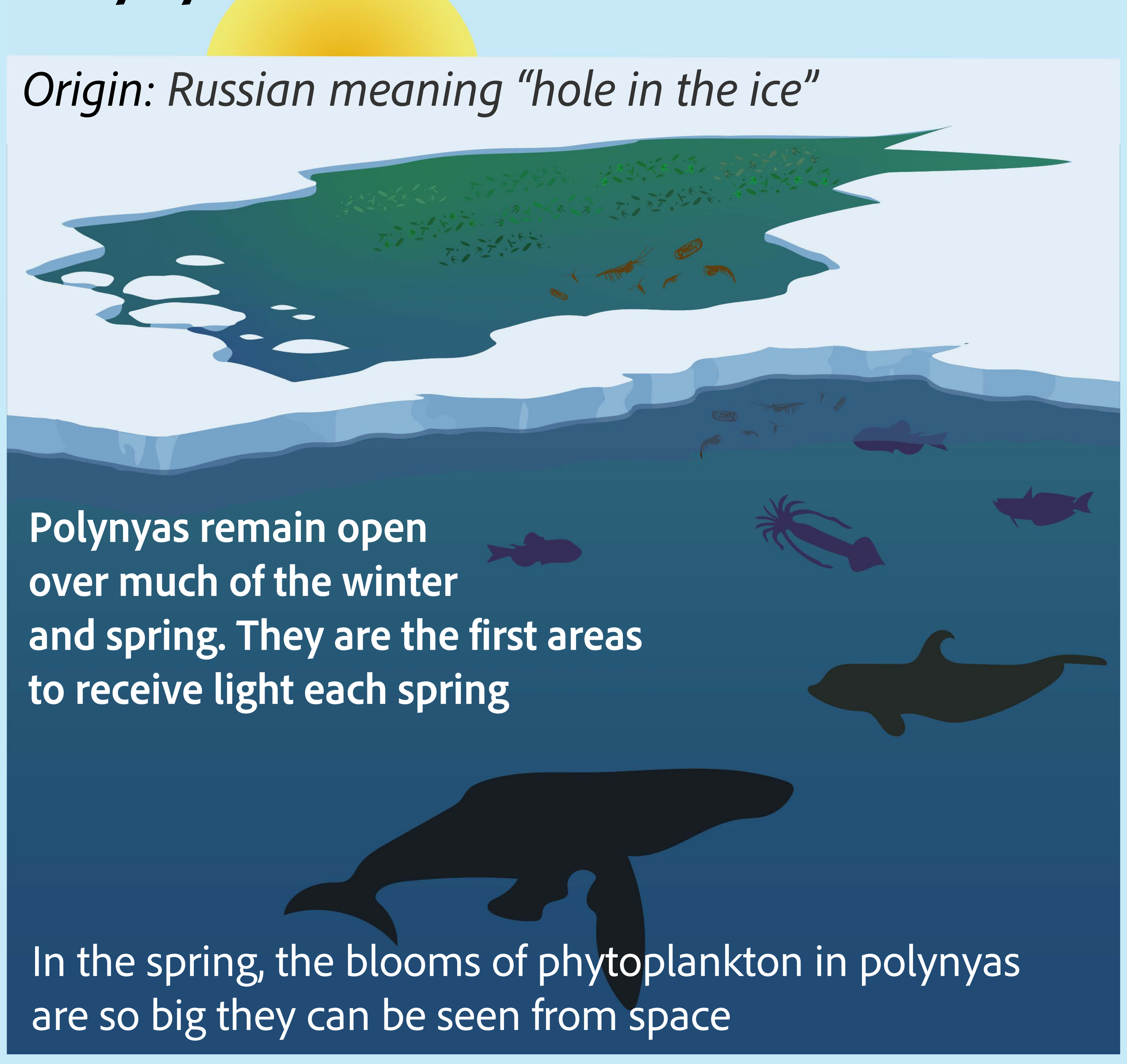
Sea Ice, Polynyas, and Productivity:
The Evolving Role of Polynyas in the Southern Ocean

Zephyr T. Sylvester[1], Kristen M. Krumhardt[2], Alice K. DuVivier[2], Laura L. Landrum[2]
[1] University of Colorado Boulder [2] National Center for Atmospheric Research (NCAR)

MOTIVATION

Polynya

Origin: Russian meaning *"hole in the ice"*



Research Question:
How will the role of polynyas as biological hotspots change in the future?

APPROACH

Community Earth System Model 2

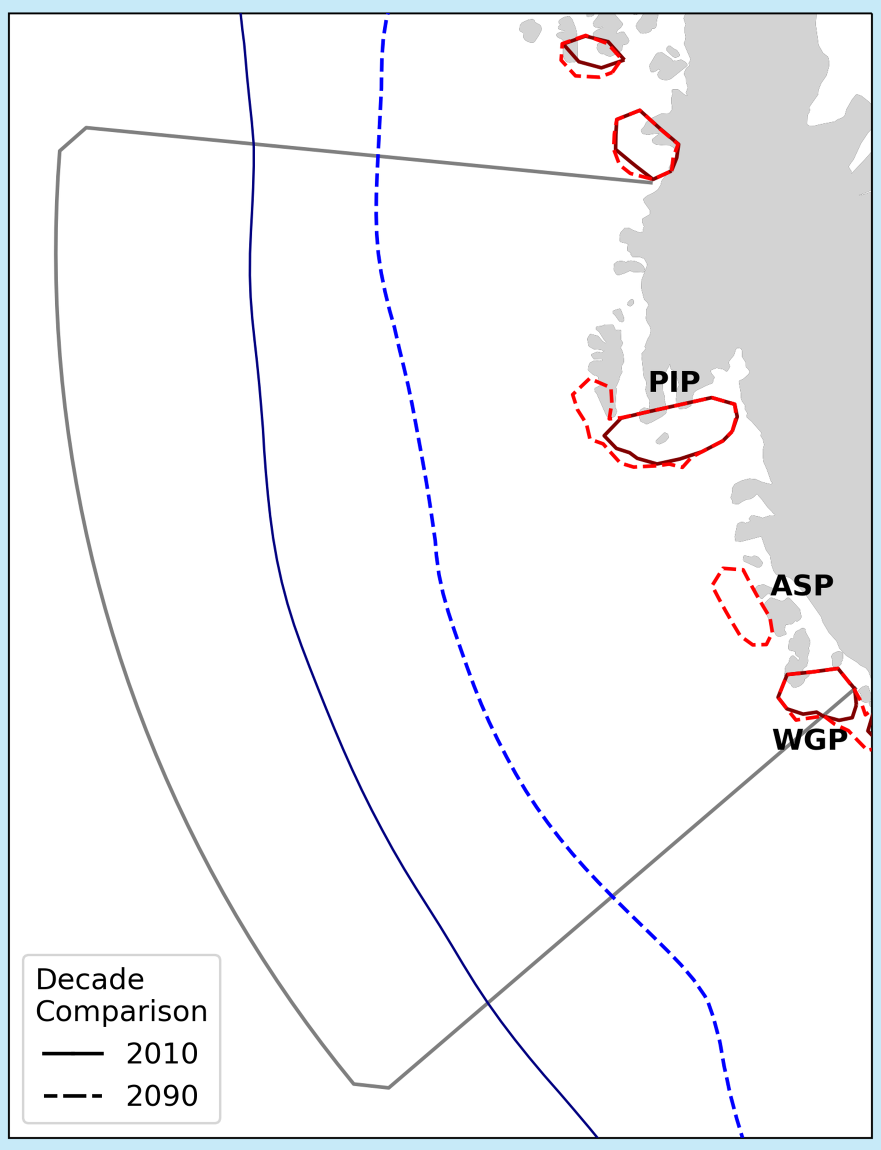
- 5 Member Ensemble
- SSP3-7.0 forcing
- MARBL Ecosystem (4P2Z)
- 1° resolution

Scan for additional details

Regional Focus: Amundsen Sea

Dynamic system: rapid sea ice loss + opposing trends to west and east

Projected favorable habitat for krill

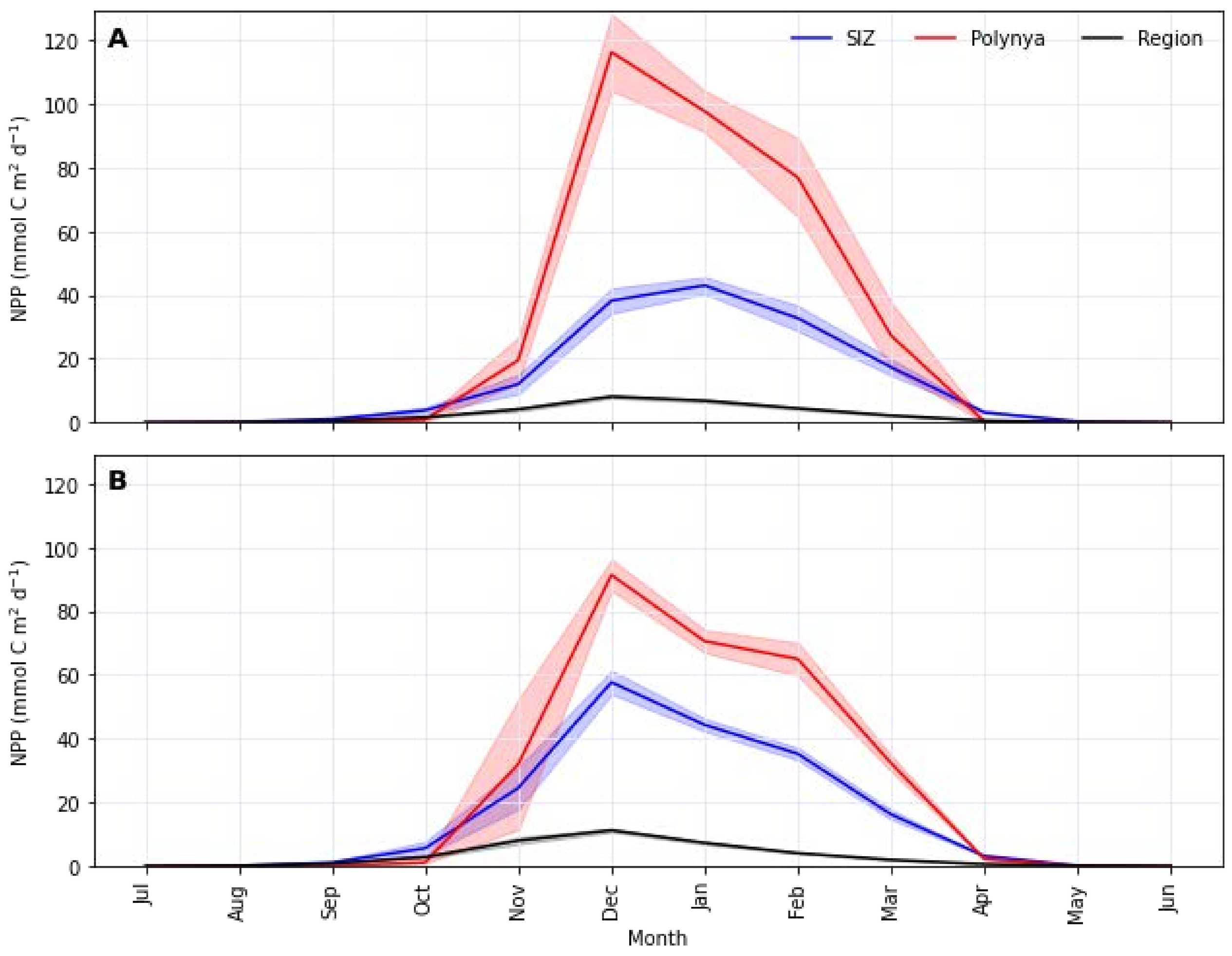


PIP: Pine Island Polynya; ASP: Amundsen Sea Polynya

Polynyas: ≥10% detection in October

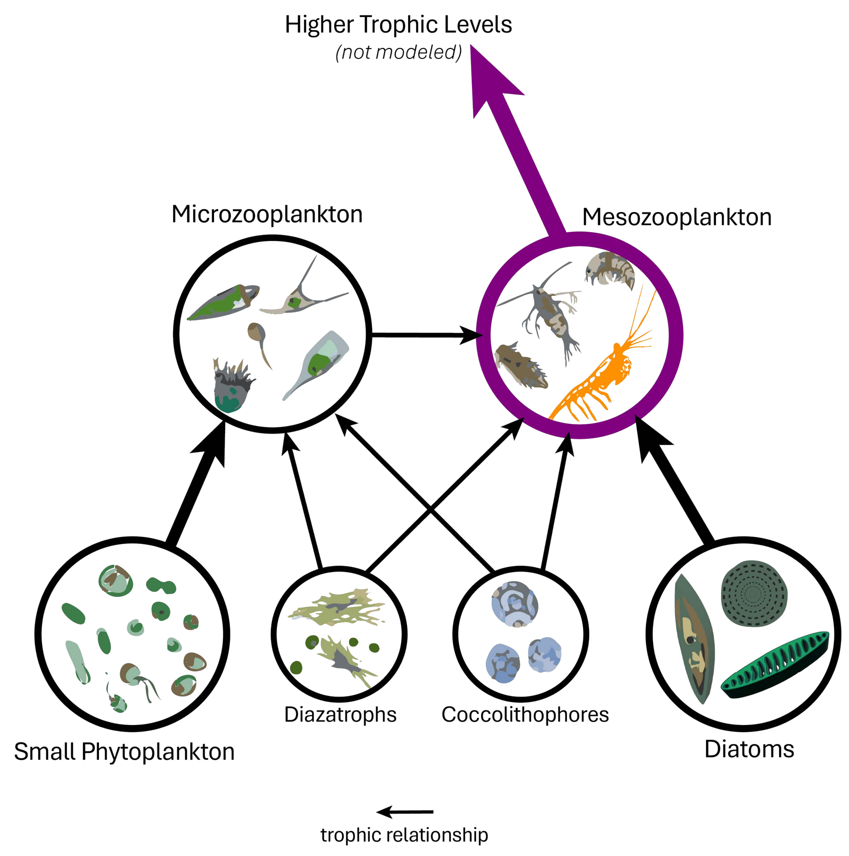
SIZ: mean winter IFRAC ≥15%

PROJECTED CHANGES



By the 2090s polynya NPP rates decline yet remain more productive than the SIZ until the winter

How could this shift in primary production impact mesozooplankton and higher-order consumers?

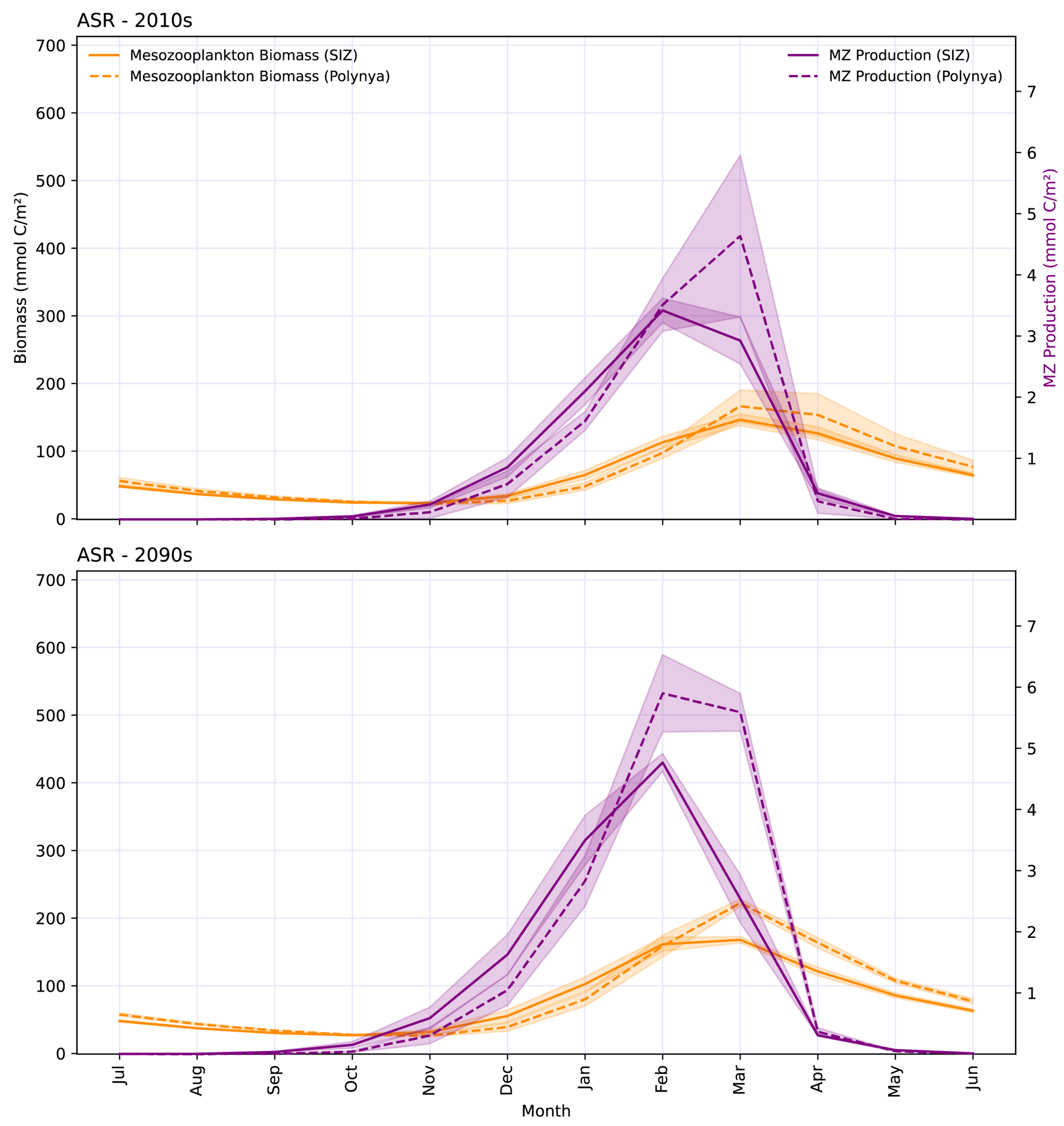


Key Findings
The NPP gap between polynyas and SIZ narrows, but the MZP gap widens

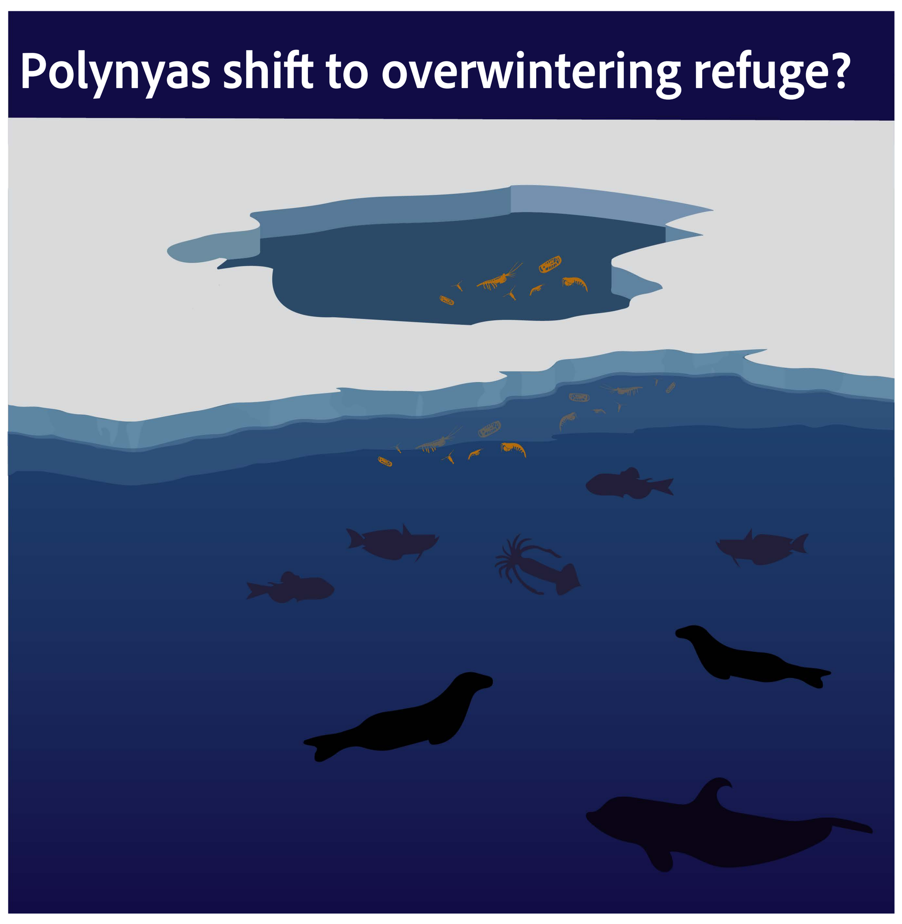
By the 2090s, seasonal MZP rates shifted: By February, MZP in polynya footprints outpaced SIZ MZP

Where diatoms persist (polynyas), MZP is sustained. Where small phytoplankton dominate (SIZ), MZP is reduced.

The resulting increased mesozooplankton biomass is higher through the winter in polynya regions compared to the SIZ



IMPLICATIONS AND FUTURE DIRECTIONS



What these projections could mean:

1) Higher MZP biomass in winter = more food for krill and their predators (seals, penguins, whales).

2) In the Amundsen Sea, the ecological significance of polynya locations may increase even as their physical distinctiveness decreases.

NASA Applied Sciences, Biodiversity and Ecological Forecasting
Grant number: 80NSSC21K1132

Contact: zephyr.sylvester@colorado.edu

Environmental Studies
UNIVERSITY OF COLORADO BOULDER

NCAR

SCAR 2026 Oslo | S25: Environmental Drivers, Human Impacts and Climate Change | Thursday